

PROJECT FLOW: Real Estate Investment Analysis

Business Objective

Help users:

- Identify **good properties to invest in**
- Understand **factors affecting property prices**

In simple:

Make smart real estate decisions using data

Business Problem

People invest without proper analysis

No clear idea:

- Which property is good
- Which area is better
- What factors increase price

So:

We solve this using data analysis

Dataset Understanding

Each row = one property

Important columns:

- Price_in_Lakhs
- Size_in_SqFt
- City
- Property_Type
- Amenities
- Public_Transport_Accessibility
- Nearby_Schools
- Nearby_Hospitals
- Parking_Space
- Age_of_Property

Data Cleaning

Handle missing values:

- Parking → fill 0
- Schools/Hospitals → fill median
- Furnished → “Unfurnished”
- Price_per_SqFt → calculate

Remove duplicates

Fix data types

Feature Engineering

Create new features:

- $\text{Price_per_SqFt} = \text{Price} / \text{Size}$
 - $\text{Age_of_Property} = \text{Current year} - \text{Year_Built}$
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Good Investment

👉 No future price used

👉 We define rules based on dataset:

Logic

A property is **Good Investment (1)** if:

- ✓ Price_per_SqFt is low (cheap)
 - ✓ Amenities are high
 - ✓ Transport is good
 - ✓ Schools nearby
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Example Rule

```
if (Price_per_SqFt < 5000 and
    Amenities == "High" and
    Public_Transport_Accessibility == "High"):
    Good_Investment = 1
else:
    Good_Investment = 0
```

Why this works?

Cheap + good facilities = high demand → good investment

EDA

◆ 1–5: Price & Size

1. Price distribution
 2. Size distribution
 3. Price per sqft by property type
 4. Size vs price
 5. Outliers
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◆ 6–10: Location

6. Avg price per sqft by state
 7. Avg price by city
 8. Property age by locality
 9. BHK distribution
 10. Top 5 costly localities
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◆ 11–14: Relationships

11. Correlation heatmap
 12. Schools vs price
 13. Hospitals vs price
 14. Furnished vs price
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◆ 16–20: Investment Factors

16. Owner type
17. Availability
18. Parking vs price

19. Amenities vs price

20. Transport vs price

Have to check for **Correlation**

How two things are related

Example

- Size $\uparrow \rightarrow$ Price $\uparrow$
Positive correlation
 - Age $\uparrow \rightarrow$ Price $\downarrow$
Negative correlation
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Deliverables

- Python
- Streamlit
- Report